

SCS 1307  
Probability & Statistics

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# Experiment and Outcome

- **A Random Experiment**

an experiment in which there is some uncertainty about the outcome that will be observed

- **Outcome**

The results of an observation of the experiment

- **Outcome space (universal set / sample space)**

Set of all possible outcomes of the experiment ( $S$ )

- **Equally Likely Outcome**

If any outcome of random experiment is as likely to occur as any other outcome then they are called equally likely outcomes.

- **Example for a Random Experiment**

Roll a die (outcomes: 1,2,3,4,5,6)

# Events

- **An Event**

A sub set of outcome space

- **Equally Likely Events**

If any event is as likely to occur as any other event then they are called equally likely events.

- **Mutually Exclusive Events**

Two events A and B are said to be mutually exclusive (or disjoint) if  $A \cap B = \emptyset$ .

- **Exhaustive events**

If two events A and B are such that  $P(A \cup B) = S$  and  $P(A \cup B) = 1$

- **An Elementary event**

An event with just one outcome

## Example

- A drawing pin is dropped twice and it is observed, each time, whether the pin is pointing upwards or downwards. Possible outcomes are:
  - \* outcome 'pin points upwards on both occasions'
  - \* outcome 'pin points downwards on both occasions'
  - \* outcome 'pin points upwards on first drop and downwards on second drop'
  - \* outcome 'pin points downwards on first drop and upwards on second drop'

# Examples

- A random experiment consists of tossing two coins and observing whether each falls Heads or tails.
- A random experiment consists of tossing three coins and observing whether each falls Heads or tails.

# Probability function

- Probability is a function , which we denote by  $P$ . Like all functions ,  $P$  has a domain which we denote by  $G$ . Since  $P$  assigns values to events,  $G$  is a set of events. A complete specification of a probability model requires us to specify
- The outcome space  $\Omega$  corresponding to the random experiment
- The domain  $G$  of the probability function
- The probability function  $P$ .

## Probability function cntd....

- A probability function  $P$ , defined on domain  $G$ , assigns to each event  $A \in G$  a unique real number  $P(A)$  in a way which satisfies the following axioms

Axiom 1:  $P(A) \geq 0$  for any  $A \in G$ .

Axiom 2:  $P(\Omega) = 1$ .

Axiom 3:  $P(A \cup B) = P(A) + P(B)$  whenever  $P(A \cap B) = \emptyset$ , for any  $A, B \in G$ .

## Some properties of P

- Property 1

$$P(A') = 1 - P(A)$$

- Property 2

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



## Some properties of P

- **Property 1**

$$P(A') = 1 - P(A)$$

Proof: Note that  $A \cap A' = \Phi$

$$\begin{aligned} \text{Therefore } P(A) + P(A') &= P(A \cup A') \text{ by axiom 3} \\ &= P(\Omega) \text{ since } A \cup A' = \Omega \\ &= 1 \text{ by axiom 2} \quad \text{hence } P(A') = 1 - P(A) \end{aligned}$$

- **Property 2**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

## Exercise 1

A random experiment consists of tossing a coin and observing whether Heads or tails results. The outcome space is  $\Omega = \{h, t\}$ . Complete the following table in a way that consistent with the axioms of probability.

| A    | $\emptyset$ | $\{h\}$ | $\{t\}$ | $\{h,t\}$ |
|------|-------------|---------|---------|-----------|
| P(A) |             | 0.4     |         |           |

## Exercise 1

A random experiment consists of tossing a coin and observing whether Heads or tails results. The outcome space is  $\Omega = \{h, t\}$ . Complete the following table in a way that consistent with the axioms of probability.

| A    | $\emptyset$ | $\{h\}$ | $\{t\}$ | $\{h,t\}$ |
|------|-------------|---------|---------|-----------|
| P(A) | 0           | 0.4     | 0.6     | 1         |

## Exercise 2

- A random experiment consists of rolling a six-sided die and observing the number of dots on the upper face. Complete the following table in a way that is consistent with the axioms of probability.

|      |             |           |         |             |             |               |                 |          |
|------|-------------|-----------|---------|-------------|-------------|---------------|-----------------|----------|
| A    | $\emptyset$ | $\{1,2\}$ | $\{3\}$ | $\{1,2,3\}$ | $\{4,5,6\}$ | $\{3,4,5,6\}$ | $\{1,2,4,5,6\}$ | $\Omega$ |
| P(A) |             |           |         | 0.5         |             |               | 0.8             |          |

## Exercise 2

- A random experiment consists of rolling a six-sided die and observing the number of dots on the upper face. Complete the following table in a way that is consistent with the axioms of probability.

|      |             |       |     |         |         |           |             |          |
|------|-------------|-------|-----|---------|---------|-----------|-------------|----------|
| A    | $\emptyset$ | {1,2} | {3} | {1,2,3} | {4,5,6} | {3,4,5,6} | {1,2,4,5,6} | $\Omega$ |
| P(A) | 0           | 0.3   | 0.2 | 0.5     | 0.5     | 0.7       | 0.8         | 1        |

# Symmetry considerations

- Physical symmetry sometimes enables us to easily assign probability values to certain events.
- Example: roll a uniform six-sided die and observe the number of dots.  
Here  $\Omega = \{1,2,3,4,5,6\}$ . Clearly each of the six elementary events is equally likely provided the die is properly constructed.
- Thus  $P(A) = n(A)/n(\Omega)$

## Example

- Toss a fair coin twice and observe the sequence of Heads and Tails.
- Here  $\Omega = \{hh, ht, th, tt\}$ . Then the probability of each elementary event is  $\frac{1}{4}$ .

# Probability values by relative frequency

- Suppose a random experiment can be repeated and one event which can be observed is  $A$ . Then if we denote  $f_r(A)$  the number of times that  $A$  occurs in  $r$  repetitions of the experiment we have the following.  $f_r(A)$  is called the *frequency of occurrence* of  $A$  in  $r$  repetitions.
- We often assign the value  $f_r(A)/r$  to  $P(A)$  if  $r$  is large.



## Example

- We drop a drawing pin 1000 times and observe that it comes to rest with the pin pointing downwards 721 times. We therefore assign  $P(D)=0.721$ , where  $D$  = event 'pin points downwards when the pin is dropped.'

## Exercise 3

- Two marbles are drawn simultaneously from a bag containing three red marbles and two blue marble. Their colours are observed. Apart from colour the marbles are indistinguishable.
  - a. Specify the outcome space
  - b. Find the probability of getting 1 red and 1 blue ball
  - c. Find the probability of getting at least 1 blue ball

## Exercise 3 - solution

- Two marbles are drawn simultaneously from a bag containing three red marbles and two blue marble. Their colours are observed. Apart from colour the marbles are indistinguishable.

a. Specify the outcome space

$$S = \{RR, RB, BB\}$$

b. Find the probability of getting 1 red and 1 blue ball

We have use extended outcome space here

$$S = \{R_1R_2, R_1R_3, R_2R_3, R_1B_1, R_2B_1, R_3B_1, R_1B_2, R_2B_2, R_3B_2, B_1B_2\} \quad \text{Answer} = 5/10$$

c. Find the probability of getting at least 1 blue ball = 7/10

## Exercise 4

- A card is drawn at random from an ordinary card pack of 52 playing cards. Find the probability that the card is;
  - a) a seven
  - b) not a seven

## Exercise 4 - solution

- A card is drawn at random from an ordinary card pack of 52 playing cards. Find the probability that the card is;
  - a) a seven  
 $= 4/52$
  - a) not a seven  
 $= 1 - 4/52 = 48/52$

## Exercise 5

- A coin and a die are thrown together. Draw a sample space diagram and find probability of obtaining
  - a head
  - a number greater than 4
  - a head and a number greater than 4
  - a head or a number greater than 4

## Exercise 5 - solution

- A coin and a die are thrown together. Draw a sample space diagram and find the probability of obtaining
  - a) a head =  $1/2$
  - b) a number greater than 4 =  $4/12$
  - c) a head and a number greater than 4 =  $2/12$
  - d) a head or a number greater than 4 =  $8/12$

## Exercise 6

- In a survey of 1500 country homes there were 1200 which had an air-conditioning unit, 900 which had a solar hot-water, while 700 had both.

A country home is selected at random, from the same area, and it is observed whether it has an air-conditioning unit and whether it has a solar hot-water unit. Give a suitable outcome space and find the probability of having

- a. only a solar hot water
- b. A solar hot water unit or an air conditioning unit



## Exercise 7

A card is drawn at random from a pack of cards. Find the probability that

1. A heart card is drawn
2. A card, which is not a face card, is drawn
3. The number on the drawn card is divisible by 3

## Exercise 7 - Solution

A card is drawn at random from a pack of cards. Find the probability that

1. A heart card is drawn =  $13/52$
2. A card, which is not a face card, is drawn =  $1 - 12/52 = 40/52$
3. The number on the drawn card is divisible by 3 =  $12/52$